

VIRAJ NISHESH DARJI

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SKILLS

Programming Languages: Python, SQL, Shell Scripting

Tools & Technology: PowerBI, Tableau, Informatica Cloud (IICS), GitHub, BitBucket, Snowflake, Databricks

Machine Learning and AI: Regression, Clustering, Text Processing, LLM, NLP, Generative AI

Data Management: Data Modeling, ETL Concepts, Data Warehousing, Data Lake, Data Quality Management, Data Governance, ERD, Data Analysis, Data Visualization, Statistical Analysis, Quality Assurance (QA) Testing, Root Cause Analysis

Agile Methodology: Scrum, Confluence, Jira, YouTrack, Project Management, Time Management

Soft Skills: Problem-solving, Critical Thinking, Adaptability, Team Collaboration, Effective Communication, Attention to Detail

PROFESSIONAL EXPERIENCE

Senior Associate Data Engineer | Egen | Fairfax, VA

Sep 2024 – Present

Heart Failure Prediction Platform for HCA Healthcare | AWS, Vertex AI, BigQuery, ML, SQL, API Development

- **Led the development of feature engineering pipelines** in BigQuery, that transformed raw structured and unstructured healthcare patient data into structured ML-ready datasets, enabling accurate heart failure prediction and impacting **5-8% of high-risk patients**.
- **Orchestrated end-to-end workflows using Cloud Composer**, automating data extraction, transformation, and storage across **50M+ records**, reducing ETL execution time by **40%**.
- **Developed a real-time API** to predict heart failure risk, reducing manual assessment time for doctors by **60%**, while logging **100% of patient interactions** in BigQuery for compliance and analytics.

Unified Data Access Model & POC for Multi-Cloud Integration | AWS, GCP BigQuery, Snowflake, Terraform, SageMaker Unified Studio

- **Led a team of three** to evaluate **SageMaker Unified Studio and AWS Lake Formation**, reducing **data access setup time by 40%** through a unified data access model.
- **Developed a Terraform-based POC** integrating **Redshift, DynamoDB, RDS, and S3**, cutting **manual provisioning efforts by 60%** and ensuring **100% compliance with governance policies** using LF-Tag-based permissions.
- **Optimized cross-platform data access**, reducing **query execution time by 30%** and streamlining authentication across **AWS, GCP BigQuery, and Snowflake**, improving operational efficiency and contributing to **\$75K+ in new business from three clients**.

Senior Data Engineer | Quantiphi Analytics | Mumbai, India

Mar 2021 – Aug 2022

Cloud Migration and Analytics Platform for Fortune 100 Insurance Company | AWS, SQL, IICS, Erwin Data Modeler, PowerBI

- **Led a 7-member team** in end-to-end agile delivery of a **\$3M project** to build an enterprise data warehouse on AWS resulting in successful migration from legacy data marts and yearly cost **savings of over \$2M**
- Designed and built Claims Data Model and delivered it with 99% accuracy and 1 month ahead of schedule, exceeding client's expectations
- Analyzed insurance data using statistical analysis and machine learning such as clustering and regression to identify regional claim patterns **increasing sales by 25%** through targeted intervention
- Developed over **30+ ETL data pipelines**, ensuring data accuracy and increased its performance by **70%** by writing complex SQL queries

Data Engineer | Quantiphi Analytics | Mumbai, India

Jul 2019 – Mar 2021

Data pipeline automation & dashboarding for global reinsurance firm | AWS, Python, Tableau

- Proposed AWS data pipeline to capture disparate P&L reports reducing manual efforts by **20 hours/week**
- **Implemented ETL in Python and deployed Tableau dashboard in 1 week securing new contract worth \$0.5M**

EDUCATION

Master of Science in Data Analytical Engineering | George Mason University | Fairfax, VA

Aug 2022 – May 2024

- **Teaching Assistant** for Fundamental of database and data mining course and Research Assistant under Dr. Plavchan on Exoplanet research

Bachelor of Engineering in Electronics and Telecommunication | University of Mumbai | India

Jul 2015 – May 2019

- Published **2 technical research paper**

CERTIFICATION

- AWS Data Analytics Specialty – a4c435a2daa8425093c0a5344ff43c04

Mar 2024 – Mar 2027

- GCP Professional Data Engineer – f88e1ad8964242e2b7c38707fcd0879a

Apr 2025 – Apr 2027

PUBLICATION

- [Automated Interpretation of Non-Destructive Evaluation Contour Maps Using Large Language Models for Bridge Condition Assessment](#)